

AI implementation: the facts

Source-verified technical supplement. Current-UI screenshot edition remains pending.

Scope and evidence

This supplement describes the current local implementation, not a deployed-service certification. The AI route, transport, client controls, recommendation logic and tests were inspected. Eight new contract tests were added; type checking and all 194 tests passed. AI tests mock the provider, so they establish application behavior, not model quality or provider compatibility.

Three generative use cases, one endpoint

The protected datapath.coach mutation accepts coach, interview and cv modes. Coaching produces learning guidance; interview mode adds a curated question and rubric; CV mode adds the learner's CV fields. All return the same message/topicIds structure. These are not three separately trained models. [S1, S3]

What is not generative AI

Roadmap construction uses authored career requirements, dependencies and learner state. Practice recommendations rank candidates with explicit weights. Quiz grading compares answers to stored keys. Spaced review uses successful days and fixed intervals of 1, 3, 7, 14 and 30 days, with failure/weak-topic overrides. These are deterministic rules, not learned mastery predictions. [S4-S6]

No claim of RAG, agents or model training

The reviewed coach call sends up to 40 unfinished topic IDs and titles as context. It contains no embedding search, vector database, retrieved lesson passages, tool definitions or training pipeline. It is bounded contextual prompting. The generic transport supports tools, but this route does not provide any. [S1, S2]

Review outcome: Describe the product as a rules-based learning platform with optional LLM assistance. Do not describe it as an autonomous tutor, verified assessor, trained recommendation model or production-validated AI system.

The earlier screenshot guide is not suitable as evidence of the current UI. No old screenshots are reused in this supplement.

Request and data flow

Browser → authenticated application route → configured LLM service → validated display.

1 Browser and admission controls

The UI enables AI actions when signed in, a provider key is reported as configured, consent is checked and the action is not pending. The request sends the entire learning state to the application server. The server requires authentication, valid state, a message of 3-2,000 characters, an allowed mode and consent=true. This boolean is not a durable consent receipt. [S1, S3]

2 Context assembled by the server

makePlan derives a plan from the submitted learner state. The first 40 unfinished topics contribute IDs and titles. Other provider fields are mode, request, role, experience, goals, roleDescription, motivation, interestedTools, expertise and assessment. A question is null or a curated interview object. Language selection appears in the system instruction. [S1]

3 Mode-specific data

Interview mode requires an ID from interviewBank(profile), and sends its question, category, skill and rubric plus the learner's answer as request. CV mode sends the complete cv object: name, summary, achievements and links. Other modes omit cv. No explicit sector, complete lesson text, practice history, proof ledger or conversation transcript is included in the provider payload. [S1, S3, S4]

4 Provider request

The server sends JSON to BUILT_IN_FORGE_API_URL plus /v1/chat/completions; when the URL is empty it defaults to https://forge.manus.im/v1/chat/completions. Authorization uses BUILT_IN_FORGE_API_KEY. The transport error text mentions OPENAI_API_KEY, but the implemented configuration reads the Forge variable. Do not infer the actual model vendor from that error text. [S2]

5 Display without automatic changes

The client shows result.message as React text and appends the titles for returned topic IDs. The endpoint does not write a new plan, change a score, update the CV or save the response to the learning database. No multi-turn transcript is sent by this UI. [S1, S3]

Controls and their limits

Structural checks reduce invalid outputs; they do not establish factual correctness.

Model and generation settings

The coach call passes no model identifier, temperature or seed. No model-version or reproducibility claim is supported. It requests `max_tokens=1800` and a strict JSON-schema response with `message:string` and `topicIds:string[]`. Provider acceptance of this request is unverified. [S1, S2]

Server-side response checks

The first choice must contain a string. JSON parsing and Zod validation then require message length at most 6,000 characters and at most five topic IDs. Every ID must belong to the exact unfinished-topic subset sent to the provider. Invalid IDs reject the response rather than being silently removed. The provider schema does not itself encode these length limits; server validation does. [S1, T1, T2]

What validation does not prove

A valid response can still contain incorrect explanations, unsupported claims or invented URLs in message. The audit demonstrates that arbitrary prose passes when structurally valid. Empty messages and repeated valid topic IDs are not explicitly rejected. Topic validation is an allowlist check, not semantic grounding or citation verification. [S1, T2]

Prompt-level guidance

The system instruction tells the model to treat user content as untrusted, avoid invented curriculum, credentials and scores, and refrain from changing the plan. These are instructions, not a demonstrated prompt-injection defense. The route has no tools to execute model commands and no automatic writeback; prose still requires human review. [S1]

Identity and state trust

Authentication identifies the caller, but the AI context is derived from validated client-submitted state. The route does not reload a canonical progress record before constructing the prompt. State validity must not be equated with independently verified skill evidence. [S1]

Expert review priority: Pin and verify a model/provider contract, test factuality and adversarial inputs, and define what the assistant may claim about learner competence.

Availability is not validation

Implementation details that affect reliability, cost and data handling.

Request limit

An in-memory Map allows 20 admitted requests per user within the preceding hour. It is local to one server process and resets on restart; it is not a distributed quota. The counter is charged before interview-ID validation and the provider call, so failed requests can consume capacity. [S1, T2]

Timeouts and retries

Each fetch attempt has a 25-second timeout. The transport permits four retries after the initial attempt for network errors, HTTP 429 and 5xx responses. It uses equal-jitter exponential backoff and Retry-After, capped at 30 seconds per delay. Other HTTP errors are not retried. The total operation can take longer than 25 seconds. There is no explicit idempotency key or overall deadline in this transport. [S2]

Error and capability behavior

A missing key returns `SERVICE_UNAVAILABLE` before provider contact. Provider, parsing and output-validation failures become `BAD_GATEWAY`. The UI combines failures into a general unavailable-or-limit message. The capability flag checks key presence, not provider health, model access or schema support. [S1, S3]

Data boundary

The server forwards the selected profile/request fields and, in CV mode, identifying CV content. The UI warns against sensitive information, but this route contains no automatic redaction. Its consent text mentions profile/request broadly; it does not enumerate all CV fields or identify the configured recipient. [S1, S3]

Unknowns that require operational evidence

Provider retention, training use, region, subprocessors, deletion behavior, billing, latency and factual accuracy cannot be established from these files. Usage may exist in a provider response, but the coach route does not record it. Do not claim zero retention, local-only processing or a proven cost per request. [S1, S2]

No live provider call was made for this review. No learner data was sent to an AI service by the audit tests.

Current UI: required coverage

Capture each feature in its actual current state, with a descriptive caption.

Capture is blocked: the browser tool cannot verify its administrator-enforced policy for localhost:3010. No alternative capture mechanism was used to bypass that restriction. The following is the required shot list, not a claim that the images have been captured.

Entry and learning

- 01 Landing page and value proposition.
- 02 Onboarding: career, industry and study-time selection.
- 03 My next lesson dashboard.
- 04 Current roadmap skill selector and an expanded lesson.
- 05 Foundations unit: worked example and independent exercise.
- 06 Quiz feedback and spaced review.

Practice Lab

- 07 Exercise browser showing path-relevant choices.
- 08 SQL editor with results and feedback.
- 09 Python task with execution output.
- 10 Excel/DAX or metric exercise.
- 11 Applied case/industry mission with dataset and self-review.

Supporting features

- 12 My progress / Proof Ledger and export preview.
- 13 Resource library.
- 14 Tools setup.
- 15 Projects and self-review.
- 16 Interview studio.
- 17 Career toolkit / CV.
- 18 Settings: saving, backup and account state.

AI-specific screenshots

- 19 Coach prompt, consent and current availability state.
- 20 Interview AI feedback controls and, if available, an actual response.
- 21 CV AI review controls and, if available, an actual response. Label disabled/unconfigured states honestly; do not invent successful responses.

Acceptance rule: use a disposable or non-sensitive demonstration profile; record the page, feature, capture date and state. Show the actual current build. A screenshot proves visible UI only, not backend correctness or model quality.

The final illustrated guide is pending these images. The previous guide must not be presented as the updated feature-by-feature screenshot edition.

How to verify this review

Local source references and validation boundaries for technical reviewers.

S1 AI endpoint and admission

server/routers.ts: limitAI (line 67), feedbackSchema (81), capabilities (146), datapath.coach (177-270).
server/_core/trpc.ts: protectedProcedure authentication middleware.

S2 Transport and configuration

server/_core/llm.ts: URL selection, request normalization, fetchWithBackoff and invokeLLM.
server/_core/env.ts: Forge endpoint/key environment mapping. No secrets are reproduced in this document.

S3 Client integration

client/src/pages/Home.tsx: askCoach (448), consent/availability controls (476), interview action (around 2103), CV action (2570), coach view (2730). These are the production page paths inspected; a generic chat showcase is not evidence of production AI behavior.

S4-S6 Deterministic learning logic

shared/learning.ts: makePlan, interviewBank, state/CV schemas and quiz construction.
shared/practice-recommendation.ts: candidate ranking. shared/review-schedule.ts: scheduling heuristic.

T1 Existing AI tests

server/interviews.generate.test.ts: authentication, consent, valid topic references, rejection of invented IDs, invalid interview question and missing provider configuration. Provider calls are mocked.

T2 New audit tests

server/ai-documentation-audit.test.ts: exact context fields, no model/tools, CV and interview payloads, server output bounds, unchecked prose, request length and per-user quota. Eight new tests passed.

Validation run: type checking and 194 tests in 24 files passed on 21 September 2026. No product-runtime code changed during this audit. No fresh live-model, database, or browser end-to-end validation is claimed.

Next verification work: provider/model integration test; factuality and prompt-injection evaluation set; consent/data-flow review; distributed quota and usage accounting; current UI screenshot capture. Source line numbers refer to the inspected working copy.